

Stride project

User Manual

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CHAPTER 1

Introduction

This manual provides a brief description of the Stride software and its features. Stride stands for **S**imulation of **t**ransmission of **i**nfectious **d**iseases. It is an agent-based modeling system for close-contact infectious disease outbreaks developed by researchers at the University of Antwerp and Hasselt University, Belgium. The simulator uses census-based synthetic populations that capture the demographic and geographic distributions, as well as detailed social networks.

Stride is an open source software. License information can be found in the project root directory in the file `LICENSE.txt`. The authors hope to make large-scale agent-based epidemic models more useful to the community.

The software was developed through collaborative efforts with numerous members at various stages of its development. In alphabetical order, the project acknowledges Philippe Beutels, Jan Broeckhove, Alexandra Cimpan, Niel Hens, Elise Kuylen, Pieter Libin, Signe Mogelmose, Astrid Sierens and Lander Willem.

More info on the project and results obtained with the software can be found in

- Willem L, Stijven S, Tijskens E, Beutels P, Hens N & Broeckhove J. (2015) Optimizing agent-based transmission models for infectious diseases, *BMC Bioinformatics*, 16:183 [1]
- Kuylen E, Stijven, S, Broeckhove, J, & Willem, L (2017) Social Contact Patterns in an Individual-based Simulator for the Transmission of Infectious Diseases, *Procedia Computer Science*, 108C:2438 [2]

Kuylen, E., Willem, L., Hens, N., & Broeckhove, J. (2019). Future ramifications of age-dependent immunity levels for measles: Explorations in an individual-based model. In *International conference on computational science* (pp. 456–467). Springer [3]

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- Kuylen, E., Willem, L., Broeckhove, J., Beutels, P., & Hens, N. (2020). Clustering of susceptible individuals within households can drive measles outbreaks: An individual-based model exploration. *Sci Rep*, 10(19645). [4]
 - Willem, L., Abrams, S., Libin, P., Coletti, P., Kuylen, E., Petrof, O., ... Hens, N. (2021). The impact of contact tracing and household bubbles on deconfinement strategies for COVID-19. *Nature Communications*, 12, 1524. [5]
 - Libin, P. J. K., Willem, L., Verstraeten, T., Torneri, A., Vanderlocht, J., & Hens, N. (2021). Assessing the feasibility and effectiveness of household-pooled universal testing to control COVID-19 epidemics. *PLoS Comp Biol*, 17(3), e1008688. [6]
 - Kuylen, E. J., Torneri, A., Willem, L., Libin, P. J. K., Abrams, S., Coletti, P., ... Hens, N. (2022). Different forms of superspreading lead to different outcomes: Heterogeneity in infectiousness and contact behavior relevant for the case of sars-cov-2. *PLOS Computational Biology*, 18(8), 1–20. [7]

Please note that functionality has been improved, extended and added since those publications.

CHAPTER 2

Software

2.1 Source code

The source code is maintained in a GitHub repository <https://github.com/lwillem/stride>.

Stride is written in C++ and is portable over Linux and Mac OSX platforms that have a sufficiently recent version of a C++ compiler. To build and install **Stride**, the following tools need to be available on the system:

- A fairly recent GNU g++ or LLVM clang++
- make
- A fairly recent CMake
- The Boost library
- LaTeX (optional, for documentation only)

2.2 Installation

To install the project, first obtain the source code by cloning the code repository to a directory . The build system for **Stride** uses the **CMake** tool. This is used to build and install the software at a high level of abstraction and almost platform independent (see <http://www.cmake.org/>). The project also include a traditional **make** front to **CMake** with For those users that do not have a working knowledge of

CMake, a front end **Makefile** has been provided that invokes the appropriate **CMake** commands. It provide the conventional targets to “build”, “install”, “test” and “clean” the project trough an invocation of make. There is one additional target “configure” to set up the **CMake/make** structure that will actually do all the work.

More details on building the software can be found in the file **INSTALL.txt** in the project root directory.

2.3 Documentation

The user manual distributed with the source code has been written in **L^AT_EX**(see www.latex-project.org).

2.4 Directory layout

The project directory structure is very systematic. Everything used to build the software is stored in the root directory:

- **main**: Code related files (sources, third party libraries and headers, ...)
 - **main/<language>**: source code, per language: cpp, python, R
 - **main/resources**: third party resources included in the project:
- **doc**: documentation files (API, manual, ...)
 - **doc/latex**: files needed to generate the user manual with Latex
- **test**: test related files (scripts, regression files, ...)

2.5 File formats

The Stride software supports different file formats:

CSV

Comma separated values, used for population input data and simulator output.
(default separator = “,”)

TXT

Text files, for the logger.

XML

Extensible Markup Language, a markup language (both human-readable and machine-readable) that defines a set of rules for encoding documents.

2.6 Testing

Unit tests are added to Stride based on Google's "gtest" framework and in the rStride codebase.

2.7 Results

The software can generates different output files:

event_log.txt

Details on transmission and/or social contacts events.

stride_log.txt

Details on the execution of the simulator (replaces the command line interface).

summary.csv (optional)

Aggregated results on the burden of disease, configuration details and timings.

3.1 Workspace

By default, **Stride** is installed in `./target/installed/` inside the project directory. This can be modified by setting the `CMAKE_INSTALL_PREFIX` on the CMake command line (see the `INSTALL.txt` file in the project root directory) or by using the CMake-LocalConfig.txt file (example file can be found in `./src/main/resources/make`).

Compilation and installation of the software creates the following files and directories:

- Binaries in directory `<install_dir>/bin`
 - *stride*: executable.
 - *gtester*: regression tests for the sequential code.
 - *rStride*: R-based simulation wrapper
- Configuration files (xml) in directory `<install_dir>/config`
 - *run_default.xml*: default configuration file for Stride.
 - ...
- Data files (csv) in directory `<project_dir>/data`
 - *contact_matrix_flanders_conditional_teachers.xml*: Social contact rates for Flanders for different locations, conditional upon presence, including teaching activities.
 - *contact_matrix_flanders_conditional_teachers_15min.xml*: Idem as above... but selecting only contact of at least 15 minutes.
 - *disease_xxx*: Disease characteristics (incubation and infectious period) for disease X.
 - *holidays_xxx*: Holiday characteristics for disease X.

- *pop_belgium600k_teacher_censushh.zip*: compressed folder with population data of 600k people for Belgium, including teaching activities.
- *immunity.xxx*: Age specific immunity profiles.
- Documentation files in directory `./target/installed/doc`
 - User manual

The install directory is also the workspace for **Stride**.

3.2 Running the simulator

From the workspace directory, the simulator can be started using the command “./bin/stride”. Arguments can be passed to the simulator on the command line:

USAGE:

```
bin/stride [-c <CONFIGURATION>] [-o <<NAME>=<VALUE>>] [--] [--version] [-h]
```

Where:

```
-c <CONFIGURATION>, --config <CONFIGURATION>
    Specifies the run configuration parameters. This should be the file name
    including the path from the workspace directory.

    Defaults to -c file=config/run\_default.xml

-o <<NAME>=<VALUE>>, --override <<NAME>=<VALUE>> (accepted multiple
    times)
    Override configuration file parameters with values provided here.

--, --ignore_rest
    Ignores the rest of the labeled arguments following this flag.

--version
    Displays version information and exits.

-h, --help
    Displays usage information and exits.
```

3.3 rStride Wrapper

An R wrapper is conceived to perform multiple runs with the C++ executable and to process model output.

4.1 Introduction

The model population consists of households, schools, workplaces and communities, which represent a group of people we define as a “ContactPool”. Social contacts can only happen within a ContactPool. When school or work is off, people stay at home and in their weekend community and can have social contacts with the other members. During other days, people are present in their household, weekday community and a possible workplace or school.

We use a *Simulator* class to organize the activities from the people in the population.

The *Rn* performs Bernoulli trials to decide whether a contact between an infectious and susceptible person leads to disease transmission. People transit through Susceptible-Exposed-Infected-Recovered states. Each *ContactPool* contains a link to its members and the *Population* stores all personal data, with *Person* objects. We distinguish different a number of types of ContactPools associated with the household, the workplace, the school, The Household is a key type because it fixed the home address of a person. The household, workplace and school clusters are handled separately from the community clusters, which are used to model general community contacts. The *Population* is a collection of *Person* objects.

Bibliography

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